

Complex Claim Verification via Human Fact-Checking Imitation with Large Language Models

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Abstract—Automatic Fact Verification (FV) aims to predict the authenticity of any claim automatically. Existing FV methods usually have satisfactory performance on simple claims. However, most of them experience performance degradation on claims with complex logic. The difficulty in reasoning and verifying complex claims stems from the presence of multi-hop logical relationships among the entities involved, as well as insufficient information. To address these challenges, this paper proposes a Human fAct-cHecking imitAtion based Fact Verification (HaHa) method, which verifies complex claims by claim decomposition, information completion and claim verification. Specifically, we utilize Large Language Models (LLMs) to decompose complex claims into several simpler sub-claims. Then, a two-step information completion module can complete insufficient information in sub-claims. After that, a FV module predicts the authenticity of each sub-claim and finally determine the authenticity of the original complex claim. Above process conforms to the human's general fact-checking process on complex claims, therefore it is interpretable and is easy to understand. With the powerful in-context learning ability of LLMs, as well as existing of other pre-trained models, HaHa only requires a small amount of training data. The experiments demonstrate HaHa outperforms several baselines on complex claims and has comparable performance on relatively simple claims.

I. INTRODUCTION

The rapid spread of misinformation online presents a threat across diverse domains, including public health [16], politics [23], and economics [2]. Because of the large quantity and rapid online dissemination of emerging information, manual FV is powerless. Consequently, automatic FV, which automatically judges the authenticity of claims, has obtained considerable research interests [12].

Existing FV models perform well in checking claims with simple and direct logic. But most of them have limited and degraded performance on claims with complex logic [1]. This is because the complex multi-hop logical relationships between entities in a complex claim make it difficult for FV

models to search for a single piece of direct evidence to prove its authenticity. In addition, the complex coreference relationships [17] or unclear semantics in a complex claim also brings challenges to FV models to fully understand it and reason about its authenticity.

Human experts also encounter complex claims when conducting FV and can not immediately judge them, either. But humans can easily leverage natural language processing capabilities to break down complex claims into several simpler sub-claims, facilitating better understanding of the claims. Subsequently, they resolve coreference relationships within each sub-claim directly, and use the relevant context in the sub-claim to query its unclear objects from external knowledge to make it clear. Finally, they can verify each sub-claim and summary the authenticity of the original complex claim.

In this paper, inspired by human expert fact-checking process, we propose a Human fAct-cHecking imitAtion based Fact Verification (HaHa) method. According to the mode of human thinking, it completes three steps, complex claims decomposition, information completion and claim verification, to determine the authenticity of the original complex claim.

Due to the fact that HaHa's FV process imitates that of human experts, users are more likely to understand and trust its verification process. Besides, the results of each intermediate step in HaHa are printed separately, which can be used by users to better check and understand the verification process and results. These advantages make the results of HaHa more interpretable. Meanwhile, it leverages the few-shot in-context learning capabilities of LLMs to accomplish tasks like claim decomposition and question generation. So there is no need of vast annotated data for fine-tuning.

HaHa is mainly tested on several subsets of a challenging dataset HOVER with complex claims [11]. The results demonstrate that in the verification of highly complex claims, HaHa surpassed several other competitive baselines.¹

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II. RELATED WORK

A. Fact Verification

Automatic FV aims to automatically predict the authenticity of claims by relevant evidence. It usually includes a series of tasks: claim detection, evidence retrieval, and claim verification. These tasks can be modeled as well researched tasks in the Natural Language Processing (NLP) community. The automatic FV can be achieved by developing a pipeline of these tasks. Among them, claim detection aims to identify claims that need to be verified for authenticity. This task is a pre-processing task and serves as the foundation for subsequent tasks. It is usually pre-processed separately in advance. Evidence retrieval aims to find information beyond the claims, such as text, tables, images, etc., to assist in verifying the claim [18]. Finally, claim verification evaluates the authenticity of claims based on retrieved evidence.

Some early works use no evidence beyond the claims themselves [6], but relying solely on the surface language patterns of claims. They fail to recognize well disguised erroneous information inside [21]. Thus more and more studies choose FV based on evidence. To better study the FV model, researchers have constructed various FV datasets containing gold evidence, such as FEVEROUS [1] and TabFact [4]. But in real scenarios, a newly emerged claim does not always have prepared gold evidence. The appropriate evidence should be retrieved from external sources. Considering the close relationship between retrieval and verification, they are usually regarded as a unified task in the latest researches.

B. Large Language Models for Reasoning

The language understanding, logical reasoning, and knowledge reserve capabilities of Large Language Models (LLMs) make them suitable for natural language processing tasks [3]. Studies find LLMs perform well even on tasks that emphasize reasoning, such as mathematical reasoning [9].

As LLM has made significant progress in various reasoning tasks, attempts to use it for FV have also begun to emerge, such as Chain-of-Thought [26], Self Ask [20] and Zero-Shot method [14], etc.. Recently, researchers have proposed more comprehensive methods that utilize LLMs in FV process. ProgramFC [19] asks LLMs to generate a program, which represents the reasoning steps guiding several sub models of FV, such as fact verification and question answering. In this method, LLM is used to clarify the overall logic of complex claims and transform a complex claim verification task into a combination of several simple subtasks. These subtasks, such as question answering and verifying simple claims, can be executed separately and synthesized to obtain the final result. FOLK [25] asks LLMs to decompose the original claim into sub-claims based on the predicate logic and generated a question for each sub-claim. Then, it calls question answering models to find the answer to each question, and compares the answer with the original information in sub-claims to obtain a judgment. Both ProgramFC and FOLK have provided inspiration for our model design, but there are

important differences among our work and them. Specifically, different from ProgramFC which encompasses all sub-tasks, such as sub-claims decomposition, query questions generation and sub-claim verification into a single program, HaHa independently processes all sub-tasks according to specific steps. Therefore, we can design appropriate models for each sub-task more flexibly and targeted. As for FOLK, after completing the decomposition of sub-claims, it verifies each sub-claim using a Q&A based approach. Differently, HaHa has steps such as semantic consistency detection and information completion before verifying the authenticity of sub-claims, making the verification of sub-claims easier and more accurate.

III. PROBLEM FORMULATION

The FV task is to determine the support or refute label Y of each input claim C based on a knowledge source E . Existing works have proposed various knowledge source settings for different scenarios. Referring to previous works [19], some knowledge source settings for FV are as follows.

Gold evidence. For each claim C , E is the set of gold evidence that can directly support or refute the claim C . It is usually manually annotated. In practical applications, manual retrieval and annotation of gold evidence require much costs and do not fully meet the requirements of fact checking.

Open-book evidence. E is a large textual corpus such as Wikipedia. The best evidence for each claim C is not provided directly. The model should firstly retrieve relative evidence pieces from the corpus E and then judge the authenticity of C based on retrieved pieces of evidence. This knowledge setting is more in line with the practical application scenarios.

Close-book evidence. The FV model does not have access to any external knowledge source, i.e., $E = \emptyset$. It needs to leverage the knowledge implicitly stored in its parameters acquired during pre-training or fine-tuning process. This setting is often explored in works that apply LLMs for FV.

IV. MODEL

Based on imitating the process of human verification of claims, the overall FV process of HaHa can be divided into several sequentially executed steps, i.e., complex claims decomposition, information completion, and claims verification. In the specific model, we design three modules corresponding to these steps. Each module has its own clear objectives and can output intermediate results, making it understandable. The overall framework of HaHa is illustrated in Fig. 1. Intermediate results like Simple sub-claims and Questions will be saved in files for manual analysis and inspection.

With complex and multi-hop relationships in complex claims, it is difficult to directly verify them. In order to facilitate the verification, HaHa first decomposes a complex claim C based on first-order predicate logic and checks the semantic entailment. Thus we obtain a set of simpler sub-claims $S = \{s_1, s_2, \dots, s_n\}$ that are semantically consistent with C , and the logical operation relationship γ between the sub-claims. There are also some coreference and unclear objects in the decomposed sub-claims that may affect

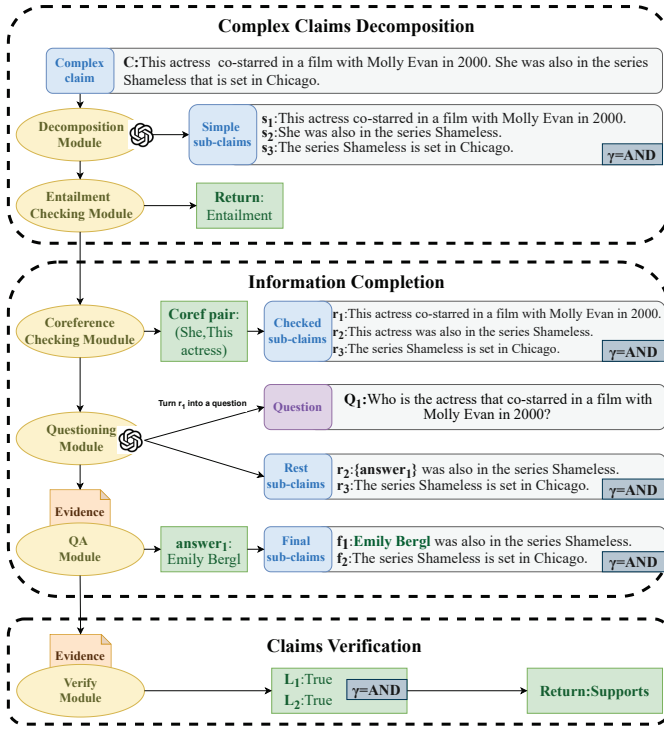


Fig. 1. The framework of the model

semantic understanding and authenticity verification of FV models. The information completion module in HaHa will eliminate semantically ambiguous coreference objects and unclear objects through coreference resolution and automatic Q&A, respectively, to obtain a set of semantically clear sub-claims $F = \{f_1, f_2, \dots, f_n\}$, that can be verify directly. Finally, the verification module verifies each simple sub-claim $f_1, f_2, \dots, f_n$ to get a support or refute judgment, i.e., $L_1, L_2, \dots, L_n$, and combine them through logical operation γ to obtain the authenticity label of the original complex claim. During the sequential execution of above steps, HaHa uses LLMs and appropriate pretrained models to complete the task.

A. Complex Claims Decomposition

A complex claim C contains multiple entities and descriptions of the various complex logical relationships that occur between them. It may be difficult for the models to understand the complex semantic and logic relationships in complex claims. They are also difficult to verify since it need to associate multiple pieces of evidence. Therefore, decomposing a complex claim into logically simple and semantically consistent sub-claims is a feasible approach. This is also consistent with the common strategy of humans to verify complex claims.

In logic, a compound proposition is composed of simple propositions and logical operators. Similarly, given a complex claim, we can decompose it into multiple simple claims composed of first-order predicate logic and some logical operators. Formally, assuming a complex claim C has multiple entities $e_1, e_2, \dots, e_{m_1}$ and predicates $v_1, v_2, \dots, v_{m_2}$, where

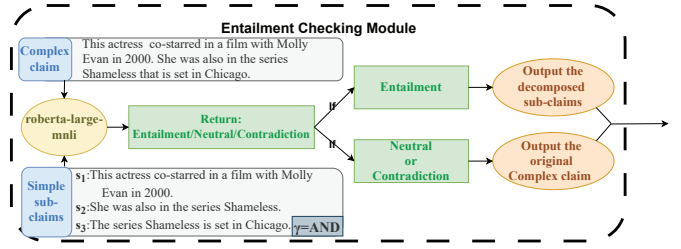


Fig. 2. The entailment checking module

each predicate and its entities form a unary predicate logic $v_k(e_q)$ or a binary predicate logic $v_k(e_q, e_p)$, then the claim can be represented as a combination of all predicate logic and logical operator, i.e., $C = v_1() \gamma v_2() \gamma \dots \gamma v_n()$, where γ represents a logical operator, like AND, OR, NAND, etc. Each first-order logical predicate v_k represents a simple sub-claim s_k . The authenticity of the complex claim can be obtained through logical operators and the authenticity of sub-claims.

The combination of first-order predicate logic and logical operators is a symbolic language. Recent works have found that LLMs can convert text input into symbolic language, such as formal language [13] or Python codes [7]. Inspired by them, we utilize the natural language comprehension ability of LLMs, to translate complex claims into a combined of first-order predicate logic and logical operators. And LLMs can further generate simple sub-claim based on each first-order predicate logic.

Formally and understandably, the decomposition task can be expressed as a logical expression $C \rightarrow s_1 \gamma s_2 \gamma \dots \gamma s_n$, and we further represent its result as a set of simple claims $S = \{s_1, s_2, \dots, s_n\}$ and a logical operator γ . Due to the complexity of this task, we adopt an in-context learning strategy to enable the LLM to better complete it. We provide the LLM with the prompt composed of task descriptions and a small number of decomposition samples, allowing the LLM to learn from them and then complete the decomposition task.

Due to the limited generalization ability of LLMs, the complex claim may not always be decomposed correctly. We analyzed and found that errors usually come from LLMs mis-understand or ignore some relationships expressed by appositives or prepositions. To solve potential semantic deviations, we set up a semantic entailment checking module. The roberta-large-mnli model [15], a roberta-large model fine-tuned on a multi-genre NLI dataset, is adopted in HaHa. It can be directly used for zero-shot text semantic consistency checking tasks and has excellent performance. We concatenate the original complex claim and the logical expression of the decomposed sub-claims together as the input, and possible outputs are "entailment", "neutral", and "contradiction". When the result is "entailment", we consider that the decomposition result is semantically consistent with the original claim. Otherwise, the decomposition result probably has semantic deviation.

Semantic deviations in the decomposition can cause deviations in the authenticity prediction. Therefore, it is meaning-

less to perform subsequent steps on simple sub-claims with semantic deviations. In theory, we can modify the prompt and repeat the decomposition until the entailment checking module outputs a "entailment". But the repetitive decomposition may increase the cost of LLM. Thus, we adopted a simpler solution in the experiment, discarding the decomposition result when there are semantic deviations, and directly treating the original complex claim as the only sub-claim for subsequent steps. The detailed process is illustrated in Fig. 2. Better solution for semantic deviations is our further research direction.

B. Information Completion

After the decomposition, the simple sub-claims $s_1, s_2, \dots, s_n$ may contain some coreference and unclear objects that affect semantic understanding. Coreference objects make sub-claims still interrelated, while unclear objects refer to entities that are described through context in a sub-claim but are not explicitly specified. These two kinds of objects make it hard for models to independently and directly verify the authenticity of each sub-claim. Specifically, the coreference objects refers to the presence of a coreferential pronoun in certain sub-claim, and the entity it refers to may be in another sub-claim, making it impossible to determine the referred entity based solely on that sub-claim. For example, in Fig.1, "She" in s_2 corresponds to "This actress" in s_1 . The unclear objects refers to some entities that are not directly and clearly provided, but need to be derived based on relevant knowledge, such as "This actress" in the s_1 itself. Both types of problems essentially lack semantically enough entity information, so a information completion module is designed to address these issues. It is also understandable since the method fits the human mind.

The coreference objects problem can be addressed by coreference resolution technology. Coreference resolution models aim to recognize objects with coreference relationships in sentences. We use the F-coref [17] as the coreference resolution model, which provides fast, accurate, and easy-to-use coreference resolution functionality. Input $s_1, s_2, \dots, s_n$ into the model, the output are some sets of objects with coreference relationships. Formally they can be represented as $(w_1, w_{11}, w_{12}, \dots, w_{1r}), (w_2, w_{21}, w_{22}, \dots, w_{2s}), \dots$. The first object in each set represents the entity, and the rest are mentions referring to such entity. The model further modifies the sub-claims based on these sets to eliminate coreferences in the sub-claims. Then the checked sub-claims set $R = \{r_1, r_2, \dots, r_n\}$ without coreference objects is got.

Some entity objects are not directly declared in the claim, but are indirectly referred to through relevant description, so called unclear objects. The decomposition and coreference resolution can not eliminate unclear objects. To solve this problem, we draw inspiration from human thinking that we use relevant description as clues to generate relevant questions, and extract the referred entities from knowledge source by answering the questions with the help of evidence. Specifically, for each claim r_i in set $R = \{r_1, r_2, \dots, r_n\}$, the LLM will judge whether it has a unclear object. If so, it generates a question Q_k to query it based on the contextual information,

and then search for answers from retrieved evidence pieces. The searched answers are adopted to replace the unclear object in rest sub-claims, so as to complete the information of the unclear objects. For example in Fig. 1, simple sub-claim r_1 has an unclear object "This actress". A question based on context describing it is generated, i.e., " $answer_1 = \text{Question('Who is the actress that co-starred in a film with Molly Evan in 2000?')}$ " and the answer " $answer_1 = \text{'Emily Bergl'}$ " is drawn from the evidence. After that we can use the $answer_1$ to replace "This actress" in r_2 . The sub-claim that provides context description to generate query question does not need to be further verified, and the rest sub-claims are now transformed into a logically simple and semantically clear set of sub-claims $F = \{f_1, f_2, \dots, f_n\}$.

The generating process also requires strong semantic and syntactic analysis capabilities, so LLMs is used again. For question answering process, we use the flan-t5-xl model [5]. It has achieved advanced zero/few-shot performance on various NLP tasks including question answering. The input prompt for the question answering sub-task is set as:

*Given evidence: {evidence}
Question: {question}
The answer is:*

C. Claim Verification

After complex claims decomposition and information completion, the model has obtained simple and clear sub-claims, together with logical operator between them. These sub-claims can be directly verified by calling verification models. For each f_k in $F = \{f_1, f_2, \dots, f_n\}$, flan-t5-xl is also adopted to predict the authenticity label L_k . The input prompt for the verification sub-task is set as:

*Given evidence: {evidence}
Based on the above information, is it true that {claim}?
True or false? The answer is:*

Finally, logical operators γ is adopted to integrate the labels of all sub-claims to yield the final support or refute label Y .

V. EXPERIMENTS

A. Datasets

We evaluated our model on datasets HOVER [11]. It is a widely used dataset that contains complex claims requiring multi-step reasoning and can best reflect the effect of each module. We use its validation sets for test because their test sets have not been released in public. HOVER includes claims that require evidence pieces from multiple Wikipedia articles for integration and reasoning. Based on the evidence required for verifying claims and the logical complexity, the validation set can be divided into three subsets: 1126 2-hop claims, 1835 3-hop claims, and 1039 four-hop claims. Usually, the more hops required, the greater the difficulty of verification.

B. Baselines

We compare the performance of HaHa with several baselines and categorize them into the following categories.

TABLE I
THE MACRO-F1 SCORE OF HAHA AND BASELINES ON HOVER.

Models	HOVER-2hop	HOVER-3hop	HOVER-4hop
BERT-FC [22]	50.68	49.86	48.57
LisT5 [10]	52.56	51.89	50.46
MultiVerS [24]	60.17	52.55	51.86
DebertaV3-NLI [8]	68.72	60.76	56.00
Direct-LLM	65.07	56.63	57.27
ProgramFC [19]	70.55	60.38	56.81
FOLK [25]	66.26	54.80	60.35
HaHa	66.57	60.87	61.61

Pretrained models. BERT-FC [22] is a pretrained model based on BERT for fact verification, while LisT5 [10] uses the T5 model for verification.

Fine-tuned models. MultiVerS [24] jointly encodes the claim and evidence document, retrieves the evidence and predicts the label through multitasking. DebertaV3-NLI [8] is the result of fine-tuning the DebertaV3 model on FEVER and some other NLI datasets.

Methods based on LLM. The Direct-LLM method directly provides evidence and the claim to a LLM for verification. ProgramFC [19] calls the LLM to generate the program for verifying the complex claim, and uses the program to guide the sub-models to run step-by-step. FOLK [25] makes use of LLMs to decompose claims and further perform question answering to determine the authenticity labels.

C. Detailed Settings

When testing HaHa, we use the openai-api to call gpt-3.5-turbo [3] as the LLM (the date of the response retrieval is around 2024.3), use flan-t5-xl [5] as the question answering and verification model, use F-coref [17] as the coreference resolution model, and use roberta-large-mnli [15], [27] as the entailment checking model. We choose these models according to the difficulty of each task and the cost, ability and advantages of models. All comparative baselines are run with the default settings as specified in their respective papers.

D. Main Results

We use macro-F1 to evaluate the performance of models. We measure their performance on 2, 3, and 4 hops claims separately. Results are displayed separately in Table 1.

The experimental results show that HaHa is superior to most FV baselines, especially when the claims are more complex. When comparing each multi-hop subset separately, HaHa falls behind DebertaV3-NLI and ProgramFC in the 2-hop scenario. However, when the complexity increases to 3-hop and 4-hop, HaHa outperforms all other baselines, demonstrating its better generalizability and adaptability to complex claims.

Specifically, the performance of models decreases with increasing hops overall, which confirms that more complex claims are more difficult to be verified. The performance decrease of HaHa is less, thus achieving performance inversion at more hops. The significant advantages of ProgramFC in 2-hop subset may stem from that it has less steps to perform verification, and the 2-hop claims are not so complex that

TABLE II
THE MACRO-F1 SCORE OF HAHA AND ITS VARIANTS WITH OPEN-BOOK.

Models	2hop	3hop	4hop
HaHa (use Q&A for information completion)	66.57	60.87	61.61
Use Q&A for direct verification	68.67	53.25	48.35
HaHa without entailment checking	64.12	59.56	57.50

TABLE III
THE MACRO-F1 SCORE OF HAHA WITH DIFFERENT EVIDENCE SOURCE SETTINGS.

Models	2hop	3hop	4hop
HaHa with gold evidence	70.99	62.71	63.69
HaHa with open-book evidence	66.57	60.87	61.61
HaHa with close-book evidence	54.19	53.63	53.70

it can be verified by relatively simple steps, while there are more steps in HaHa which are able to handle more complex claims, but also increase the possibility of error accumulation. DebertaV3-NLI also perform well here, proving the fine-tuning helps it handle FV tasks. But as the number of hops increases, the FV steps in ProgramFC and the fine-tuning of DebertaV3-NLI become insufficient, while the effect of more detailed steps in HaHa emerges. The better performance of HaHa in 3-hop and 4-hop subsets stem from the fact that there are more coreference objects and unclear objects in more complex claims, while the information completion module of HaHa can resolve the coreference objects and replace the unclear objects with concrete entities, so the final sub-claims are more clear to be verified more correctly.

HaHa does not require setting any parameters, so it should have low sensitivity. But because of the diversity of LLMs' generate result, there are still specificity each time.

E. Discussion

Q&A for Information completion vs. Q&A for Verification. In many models, Q&A is adopted for direct verification, where the authenticity prediction of the claim is obtained by comparing the answer to the question with information from the original claim. But the Q&A in information completion module of HaHa is different. The Q&A here is not for verification, but extracting concrete entities from evidence to replace the unclear objects, making the claim clear.

We design a variant of HaHa by removing the Q&A in information completion and using Q&A as direct verification method. Another variant is HaHa without entailment checking. The comparison of their performance on open-book setting is illustrated in Table 2. It shows the removal of Q&A in information completion greatly reduces the performance of 3 and 4-hop claims verification. This is because the more complex claims may have more unclear objects that can degenerate the verification. However, in 2-hop, the results have slightly increased. This is consistent with our conjecture that a 2-hop claim is simple enough to be verified directly, while additional operations may increase the overall probability of error accumulation. It also indicates that the removal of entailment checking results in worse performance.

Different Evidence Settings. The comparison between three evidence settings is illustrated in Table 3. It shows the open-book setting performs relatively close to gold setting, indicating that HaHa can retrieve evidence pieces good enough for verification. However, the result of close-book setting is far inferior to that of open-book, as with the absence of external knowledge source, relying solely on the knowledge of flan-t5-xl itself can not provide enough support.

VI. CONCLUSION

In this paper, a Human fAct-cHecking imitAtion based Fact Verification method for complex claims called HaHa is proposed by imitating the human’s general fact-checking process on complex claims. It first decomposes complex claims into several simpler sub-claims to simplify their complex logical relationships by utilizing the in-context learning ability of LLMs. Then a two-step information completion module is designed to complete two types of important information that may be missing in each sub-claim. Finally, the authenticity of the original complex claims is obtained by integrating the authenticity of each sub-claim. The experiments on a challenging FV datasets with complex claims demonstrate the superiority of HaHa over several FV baselines. The proposed method still faces some limitations that the generalization ability of LLM in decomposition tasks is sometimes unsatisfactory, leading to semantic deviation issues, and the retrieved evidence in open-book settings is sometimes unrelated to claim verification. The resolution of these limitations is our future research direction.

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REFERENCES

- [1] Aly, R., Guo, Z., Schlichtkrull, M.S., Thorne, J., Vlachos, A., Christodoulopoulos, C., Cocarascu, O., Mittal, A.: Feverous: Fact extraction and verification over unstructured and structured information. In: Thirty-fifth Conference on Neural Information Processing Systems Datasets and Benchmarks Track (Round 1) (2021)
- [2] Assenza, T., Collard, F., Fève, P., Huber, S.J.: From buzz to bust: How fake news shapes the business cycle. Tech. rep., ECONtribute Discussion Paper (2024)
- [3] Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J.D., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., et al.: Language models are few-shot learners. *Advances in neural information processing systems* **33**, 1877–1901 (2020)
- [4] Chen, W., Wang, H., Chen, J., Zhang, Y., Wang, H., Li, S., Zhou, X., Wang, W.Y.: Tabfact: A large-scale dataset for table-based fact verification. In: International Conference on Learning Representations (2019)
- [5] Chung, H.W., Hou, L., Longpre, S., Zoph, B., Tay, Y., Fedus, W., Li, Y., Wang, X., Dehghani, M., Brahma, S., et al.: Scaling instruction-finetuned language models. *Journal of Machine Learning Research* **25**(70), 1–53 (2024)
- [6] Dungs, S., Aker, A., Fuhr, N., Bontcheva, K.: Can rumour stance alone predict veracity? In: Proceedings of the 27th international conference on computational linguistics. pp. 3360–3370 (2018)
- [7] Gao, L., Madaan, A., Zhou, S., Alon, U., Liu, P., Yang, Y., Callan, J., Neubig, G.: Pal: Program-aided language models. In: International Conference on Machine Learning. pp. 10764–10799. PMLR (2023)
- [8] He, P., Gao, J., Chen, W.: Debertav3: Improving deberta using electra-style pre-training with gradient-disentangled embedding sharing. In: The Eleventh International Conference on Learning Representations (2022)
- [9] Imani, S., Du, L., Shrivastava, H.: Mathprompter: Mathematical reasoning using large language models. In: Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 5: Industry Track). pp. 37–42 (2023)
- [10] Jiang, K., Pradeep, R., Lin, J.: Exploring listwise evidence reasoning with t5 for fact verification. In: Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 2: Short Papers). pp. 402–410 (2021)
- [11] Jiang, Y., Bordia, S., Zhong, Z., Dognin, C., Singh, M., Bansal, M.: Hover: A dataset for many-hop fact extraction and claim verification. In: Findings of the Association for Computational Linguistics: EMNLP 2020. pp. 3441–3460 (2020)
- [12] Jin, Y., Wang, X., Yang, R., Sun, Y., Wang, W., Liao, H., Xie, X.: Towards fine-grained reasoning for fake news detection. In: Proceedings of the AAAI Conference on Artificial Intelligence. vol. 36, pp. 5746–5754 (2022)
- [13] Kim, Y.: Sequence-to-sequence learning with latent neural grammars. *Advances in Neural Information Processing Systems* **34**, 26302–26317 (2021)
- [14] Kojima, T., Gu, S.S., Reid, M., Matsuo, Y., Iwasawa, Y.: Large language models are zero-shot reasoners. *Advances in neural information processing systems* **35**, 22199–22213 (2022)
- [15] Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., Levy, O., Lewis, M., Zettlemoyer, L., Stoyanov, V.: Roberta: A robustly optimized bert pretraining approach. arXiv preprint arXiv:1907.11692 (2019)
- [16] Okoro, Y.O., Ayo-Farai, O., Maduka, C.P., Okongwu, C.C., Sodamade, O.T.: A review of health misinformation on digital platforms: challenges and countermeasures. *International journal of applied research in social sciences* **6**(1), 23–36 (2024)
- [17] Otmazgin, S., Cattani, A., Goldberg, Y.: F-coref: Fast, accurate and easy to use coreference resolution. In: Proceedings of the 2nd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the 12th International Joint Conference on Natural Language Processing: System Demonstrations. pp. 48–56 (2022)
- [18] Ousidhoum, N., Yuan, Z., Vlachos, A.: Varifocal question generation for fact-checking. In: Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing. pp. 2532–2544 (2022)
- [19] Pan, L., Wu, X., Lu, X., Luu, A.T., Wang, W.Y., Kan, M.Y., Nakov, P.: Fact-checking complex claims with program-guided reasoning. In: The 61st Annual Meeting Of The Association For Computational Linguistics (2023)
- [20] Press, O., Zhang, M., Min, S., Schmidt, L., Smith, N.A., Lewis, M.: Measuring and narrowing the compositionality gap in language models. In: The 2023 Conference on Empirical Methods in Natural Language Processing (2023)
- [21] Schuster, T., Schuster, R., Shah, D.J., Barzilay, R.: The limitations of stylometry for detecting machine-generated fake news. *Computational Linguistics* **46**(2), 499–510 (2020)
- [22] Soleimani, A., Monz, C., Worring, M.: Bert for evidence retrieval and claim verification. In: Advances in Information Retrieval: 42nd European Conference on IR Research, ECIR 2020, Lisbon, Portugal, April 14–17, 2020, Proceedings, Part II 42. pp. 359–366. Springer (2020)
- [23] Stachofsky, J., Schaupp, L.C., Crossler, R.E.: Measuring the effect of political alignment, platforms, and fake news consumption on voter concern for election processes. *Government Information Quarterly* **40**(3), 101810 (2023)
- [24] Wadden, D., Lo, K., Wang, L.L., Cohan, A., Beltagy, I., Hajishirzi, H.: Multivers: Improving scientific claim verification with weak supervision and full-document context. In: Findings of the Association for Computational Linguistics: NAACL 2022. pp. 61–76 (2022)
- [25] Wang, H., Shu, K.: Explainable claim verification via knowledge-grounded reasoning with large language models. In: The 2023 Conference on Empirical Methods in Natural Language Processing (2023)
- [26] Wei, J., Wang, X., Schuurmans, D., Bosma, M., Xia, F., Chi, E., Le, Q.V., Zhou, D., et al.: Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems* **35**, 24824–24837 (2022)
- [27] Williams, A., Nangia, N., Bowman, S.R.: A broad-coverage challenge corpus for sentence understanding through inference. In: Proceedings of NAACL-HLT. pp. 1112–1122 (2018)